

CASSIA EDWARDS

Email: s2025873@ed.ac.uk ◇ Edinburgh, UK

RESEARCH INTERESTS

Broadly I am interested in pure mathematics with connections to theoretical physics. More specifically, homotopy theory and higher algebraic structures in connection with quantum field theory, in particular factorisation algebras. My work predominately involves ∞ -categories.

EDUCATION

EPSRC Centre for Doctoral Training in Algebra, Geometry, and Quantum Fields

University of Edinburgh Lead

Autumn 2025 - (Summer 2029)

PhD with Integrated Study in Algebra, Geometry and Mathematical Physics

Supervisor: Clark Barwick

University of Edinburgh

Autumn 2020 - Summer 2024

MMath Mathematics

First Class Honours

Original degree path Joint Honours Mathematics and Physics BSc. Transferred to Integrated Master's Mathematics MMath into fourth programme year (September 2022). Entered directly into second year of degree programme.

RESEARCH EXPERIENCE

Families of Circles

Supervisor: Prof. Tony Carbery

Sep 2023 - March 2024

Master's Dissertation

Kummer Quartic Surfaces

Supervisor: Prof. Ivan Cheltsov

Sep 2022 - March 2023

Bachelor's Dissertation

Investigating The Quantum Anharmonic Oscillator

Supervisor: Prof. Luigi Del Debbio

July 2022 - September 2022

Summer Research Internship

EXPOSITORY TALKS

A Lightning Introduction To Operads

Intermediate Category Theory Reading Group

December 2025

University Of Edinburgh

Chiral Algebras (At Least Somewhat) Demystified

Hodge Club

March 2026

University Of Edinburgh

Motivating Factorisation Algebras in Quite a Lot of Generality

GEARS Seminar

June 2026

University of Edinburgh and University of Glasgow

TEACHING EXPERIENCE

University Tutor in Mathematics

University of Edinburgh

Autumn 2024 - Summer 2025

Full time position. Duties include facilitating the provision of small group work-shop style sessions; assignment and exam marking and delivering revision lectures.

FUNDING AND AWARDS

Martingale Foundation, PhD study funded by the Martingale Foundation, a charity that funds postgraduate research degrees on a competitive basis. *Autumn 2020 - (Summer 2029)*

Research Internship Funding, awarded by University of Edinburgh School of Physics and Astronomy to complete an extracurricular summer research project. *Summer 2022*

Scholarship, awarded by The Lords Feofees and Assistants of the Manor to students from Bridlington, UK for a maximum of three years that demonstrate academic potential and motivation. *2020 - 2023*

Access Edinburgh Scholarship, awarded by the University of Edinburgh to full-time undergraduates from widening participation backgrounds. *2020-2024*

Mathematics Award, awarded by Scarborough Sixth Form College and the Old Girls' High School Association to the highest achieving female student in mathematics in a given year. *Summer 2020*